

BIS Sahayak (Bureau of Indian Standards Assistant)

AI-Powered Quality & Safety Companion for Indian Citizens and Small Businesses

National Hackathon: Smart India Hackathon (SIH) 2026	Problem Statement ID: SIH26107 (Software)
Project Name: BIS Sahayak (Bureau of Indian Standards Assistant)	Category: GovTech / Artificial Intelligence / Public Safety
Institution: Arvind Gavali College of Engineering, Satara	Live Portal: sih2026-bis-assistant.vercel.app

1. EXECUTIVE SUMMARY: THE ENTIRE PROJECT IN 2 MINUTES

What is the Bureau of Indian Standards (BIS)?

In India, whenever you purchase an everyday product—such as a kitchen pressure cooker, electrical wiring, a bottle of packaged drinking water, or a motorcycle helmet—you will notice a small seal called the **ISI mark**. The Bureau of Indian Standards (BIS) is the Government of India's apex national standards body. Its constitutional mission is to formulate and enforce strict quality, safety, and reliability rules across manufactured goods to protect citizens and elevate national industry.

What is BIS Sahayak?

BIS Sahayak (Bureau of Indian Standards Assistant) is a free, 24/7 web-based AI assistant and product authenticity verification platform. Think of it as having an expert senior government safety inspector and a seasoned quality compliance consultant available instantly in your pocket. It digests thousands of pages of dense government gazette notifications and engineering standards, and delivers crisp, instant answers in plain, accessible English and Hindi. Furthermore, it incorporates an interactive **ISI Mark Authenticity Verifier** that empowers any citizen to verify whether the license number printed on a product is genuine or a hazardous counterfeit.

KEY HIGHLIGHTS

Core Innovations at a Glance:

- Dual-Persona Intelligence:** Seamlessly toggles between a simple, friendly mode for everyday consumers and an exact engineering mode for factory owners.
- Zero-Hallucination Architecture:** Unlike generic chatbots (like ChatGPT) that guess or invent numbers, BIS Sahayak answers strictly from pre-loaded authentic BIS standard documents.
- Live CM/L License Verifier:** Validates 7 or 8-digit license numbers against certified manufacturer registries and guides users to the official BIS Care Mobile App.
- Voice & Bilingual Accessibility:** Fully bilingual in English and Devanagari Hindi, equipped with one-tap speech synthesis (🔊) for audio assistance.

2. THE REAL-LIFE PROBLEM: WHY WAS BIS SAHAYAK BUILT?

Every single day in India, millions of consumers and factory owners make quality-related decisions. However, the critical information needed to guarantee safety has been historically trapped behind insurmountable barriers.

1. The Everyday Consumer's Dilemma (Safety Hazards & Counterfeits):

Consider a mother shopping in a local market for a pressure cooker or plastic toys for her children. She notices an ISI stamp on the box. How can she verify if that mark is genuine or a counterfeit stamp? Sub-standard pressure cookers explode under domestic kitchen heat; cheap uncertified electrical cables spark house fires; fake children's toys frequently contain toxic lead and heavy metals; and spurious bottled water spreads microbial pathogens. Most citizens do not know that genuine ISI marks must carry a 7 or 8-digit CM/L license number beneath the emblem, nor do they know what specific physical safety mechanisms to inspect before buying.

2. The Small Business & Factory Owner's Dilemma (Regulatory Friction):

India is home to over 63 million Micro, Small, and Medium Enterprises (MSMEs). When ministries issue mandatory **Quality**

3. The Institutional Gap:

THE CORE PROBLEM STATEMENT

Indian Standards protect human lives and drive industrial excellence, but they have been locked in complex legal PDFs that ordinary citizens cannot read and small businesses struggle to search.

BIS Sahayak bridges this divide by delivering an accessible, 24/7 web-based companion engineered with five key superpowers:

- **Dual-Persona Intelligence (Consumer vs. Industry):** Users switch between two dedicated personas with one tap. **Consumer Mode** provides friendly everyday language, explains what physical features to check, and warns against fake stamps. **Industry / MSME Mode** delivers exact engineering limits (such as 500 MPa yield stress, spark test voltages, routine vs. type testing regimes, and HS trade classification codes).
- **Live 4-Step ISI License Authenticity Verifier (/verify):** Allows any user to input a 7 or 8-digit CM/L license number. It validates the format, searches verified manufacturer records (including Prestige, Havells, Tata Steel, Finolex), displays registered factory locations, and flags unverified numbers with red counterfeit alerts and direct steps to file complaints on the BIS Care App.
- **Standards Directory & Slide-Out Clause Drawer (/standards):** An interactive catalog containing 21 authentic Indian Standards across Civil, Electrical, Kitchen, Consumer Safety, and Energy sectors. Users can click any citation badge inside the chat to open a side drawer with official clauses without losing their conversation.
- **100% Bilingual Support & Web Speech Audio Assistant:** Every screen, button, and AI answer is natively supported in English and Hindi (Devanagari script). With built-in Web Speech synthesis, non-literate or visually impaired users can tap the speaker icon (🔊) to listen to answers read aloud.
- **Zero-Hallucination Retrieval-Augmented Generation (RAG):** Because safety limits cannot tolerate AI errors, the system indexes authentic government standards into mathematical vector embeddings. Google Gemini 2.5 Flash is strictly restricted to generate answers using only retrieved standard clauses.

4. REAL-LIFE STORIES: HOW PEOPLE USE BIS SAHAYAK

1. Clarifies that domestic pressure cookers must be certified under **IS 2347:2017**.
2. Gives him a 3-point physical safety check: (a) Weight Valve, (b) Metallic Safety Plug, and (c) Gasket Release System.
3. Emphasizes: *'Always confirm the ISI mark has a 7 or 8-digit CM/L license number beneath it.'*
4. Ramesh clicks the speaker button (■) to hear the answer read aloud in clear Hindi.
5. He navigates to **Verify**, inputs the number on the box, and confirms the manufacturer's active certification.

STORY B: SMALL FACTORY OWNER (INDUSTRY MODE)

Scenario: Priya operates a steel rolling mill in Jalna and seeks to supply Fe 500D TMT rebars for public infrastructure tenders.
Action: She switches to **Industry / MSME Mode** on the portal.
Question: 'What are the mandatory tensile and elongation limits for Fe 500D under IS 1786?'
BIS Sahayak's Response:
• **Proof Stress (0.2% Yield):** Minimum 500.0 N/mm² (MPa)
• **Tensile Strength (UTS):** Minimum 565.0 N/mm²
• **Elongation Percentage:** Minimum 16.0% (mandatory for seismic ductility)
• **Clause Drawer:** Priya clicks the badge **[IS 1786:2008]** directly in the chat, opening an instant drawer displaying Clause 8.1 (Chemical Analysis) and Clause 9.2 (Tensile Protocols) without leaving the screen.

STORY C: PROTECTING CHILDREN FROM TOXIC TOYS

Scenario: Sunita is buying plastic toys for her 2-year-old child and wants to know if cheap uncertified toys pose risks.
BIS Sahayak's Response: Explains that under **IS 9873** and the Government Toy Quality Control Order, uncertified toys often contain dangerous heavy metals (lead, cadmium, phthalates) and pose choking hazards. It gives her visual guidelines to identify compliant toy packaging.

5. HOW IT WORKS UNDER THE HOOD (EXPLAINED SIMPLY)

Here is the five-step technical process explained in simple, easy-to-understand concepts:

Step 1: Reading & Chunking	Official BIS standards were segmented into clean, structured knowledge blocks: Standard code, Scope, Clauses, Physical & Chemical Limits, and Gazette Mandates.
Step 2: Vector Embeddings	Each clause is converted into 768-dimensional mathematical coordinates. Concepts with similar meanings are stored close together in mathematical space.
Step 3: Fast Hybrid Search (<15ms)	When a question arrives, the engine performs hybrid search combining cosine vector math with domain-aware stop-word filtering to identify exact matching clauses in under 15 milliseconds.
Step 4: AI Formulation (Gemini 2.5)	Google Gemini 2.5 Flash is invoked with strict low temperature (0.3). It is constrained to use ONLY the retrieved clauses, guaranteeing zero hallucinations.
Step 5: Interactive Presentation	The user receives a clean 2-sentence summary, expandable engineering data tables, clickable citation chips, and live speech audio playback.

ARCHITECTURE & RELIABILITY

Resilient Serverless Architecture:
• **Frontend:** Next.js 15 (React 19) and Tailwind CSS deployed globally on Vercel Edge Cloud.
• **Database:** Neon Serverless PostgreSQL with pgvector, scaling to zero idle cost.
• **Offline Resilience:** An in-memory vector index ensures continuous 100% uptime even if external databases encounter network disconnects.

6. THE STANDARDS CATALOG: 21 PRE-LOADED INDIAN STANDARDS

The current deployment includes 21 authentic Indian Standards representing critical manufacturing and consumer sectors:

Category	Standard	Product Covered	Key Safety & Quality Limits
Civil	IS 1786:2008	Fe 500D TMT Rebars	Min 500 MPa yield, min 16% elongation
Civil	IS 269:2015	Portland Cement	Compressive strength & sound setting time
Civil	IS 4985:2021	uPVC Water Pipes	Hydrostatic pressure & impact resistance
Electrical	IS 694:2010	PVC Insulated Cables	6 kV spark test, flame retardancy (FRLS)
Electrical	IS 1293:2019	Plugs & Sockets (6A/16A)	Child safety shutters, temperature rise <45K

Electrical	IS 15885:2012	LED Lighting Drivers	Thermal overload & short-circuit protection
Kitchen	IS 2347:2017	Pressure Cookers	Safety plug, weight valve, gasket release
Kitchen	IS 302-1:2024	Household Appliances	Earthing continuity, leakage current limits
Kitchen	IS 2082:2018	Electric Water Heaters	Pressure relief, non-scalding thermostat
Kitchen	IS 4250:2022	Mixer Grinders	Thermal overload cutout, double insulation
Consumer	IS 14543:2024	Packaged Drinking Water	Zero coliform, pesticide limits <0.0001 mg/L
Consumer	IS 10500:2012	Drinking Water Quality	Permissible limits for pH, TDS, heavy metals
Consumer	IS 9873:2019	Safety of Toys	No swallowable small parts; zero toxic lead
Consumer	IS 16018:2012	Fire Extinguishers	Discharge duration, hydraulic burst test
Industrial	IS 3196:2022	LPG Gas Cylinders	Hydrostatic stretch test, burst >5.1 MPa
Industrial	IS 4151:2020	Motorcycle Helmets	Impact absorption, chin strap retention
Industrial	IS 15298:2016	Safety Footwear	200-Joule toe cap impact, slip resistance
Energy	IS 16046:2018	Lithium-Ion Batteries	Overcharge protection, thermal abuse check
Energy	IS 7318:2020	Solar PV Inverters	Anti-islanding protection, THD <5%

7. WHY BIS SAHAYAK WINS: COMPARISON WITH OTHER METHODS

Criterion	Government Gazette PDFs	General AI (ChatGPT)	BIS Sahayak (Our Innovation)
Accessibility	Unsearchable, heavy legal PDFs.	Conversational, lacks Indian context.	Dual Personas: Simple layman or deep engineering.
Accuracy & Risk	Accurate but hard to extract.	Hallucinates: Invents false MPa or volts.	Zero-Hallucination: Bound to authentic BIS text.
ISI Verification	Hidden behind complex portals.	Cannot verify 7/8-digit CM/L numbers.	Live 4-Step Verifier: Format check + BIS Care guide.
Voice & Audio	No speech synthesis.	Requires paid upgrades.	Web Speech: Instant voice in Hindi and English.
Cost & Barrier	Needs expensive lawyers/consultants.	Subscription fees (\$20/month).	100% Free & Open: Works in any mobile browser.

8. NATIONAL IMPACT: BENEFITS FOR INDIA

- 1. Protecting 1.4 Billion Citizens (Social Impact & Life Safety): Empowers everyday families to detect dangerous counterfeit products. An exploding pressure cooker, a faulty geyser, or spurious building wires can destroy lives. BIS Sahayak puts safety knowledge directly in citizens' hands, democratizing consumer protection.
- 2. Empowering 63 Million Small Businesses (Make in India & Ease of Doing Business): Drastically reduces regulatory lookup times from days to seconds. Small manufacturers can verify exact testing tolerances, routine test frequencies, and mandatory QCO gazette dates without paying expensive consulting retainers.
- 3. Supporting the Bureau of Indian Standards (Institutional Adoption): Builds public awareness and trust in the ISI mark, while directly boosting downloads and grievance filings on the official BIS Care Mobile App.
- 4. Ethical & Sustainable Computing: The serverless cloud infrastructure consumes energy strictly on demand, minimizing cloud carbon footprints. No user personal data is captured, stored, or monetized.

9. TESTING, QUALITY ASSURANCE & SCALABILITY ROADMAP

Automated Test Suite: The system includes 15 automated test suites executed via Vitest. These verify sub-20ms vector retrieval, CM/L license pattern detection (catching invalid test numbers like 11111111), and complete parity in English/Hindi

translation keys.

Future Roadmap:

- **Phase 1 (Completed Live Prototype):** 21 core standards indexed, dual personas, 4-step ISI verifier, Web Speech voice assistant, and live cloud deployment on Vercel.
- **Phase 2 (Next 6 Months):** Scaling from 21 to all 20,000+ Indian Standards; direct live API sync with the official e-BIS / Manakonline portal for real-time license verification.
- **Phase 3 (Next 12 Months):** WhatsApp Chatbot integration for rural low-bandwidth areas; Mobile Camera OCR to scan product boxes and verify ISI marks automatically; support for 10 regional Indian languages.

10. QUICK REFERENCE, ACCESS LINKS & TEAM INFORMATION

Resource	Location / URL	Description
Live Web Application	https://sih2026-bis-assistant.vercel.app	Accessible 24/7 on any mobile or desktop browser
GitHub Source Code	https://github.com/PhoenixOP231/BIS_Sahayak_SIH	Complete open-source repository with automated tests
SIH Problem Statement	SIH26107 (Smart India Hackathon 2026)	AI-Powered Assistant for Indian Standards & BIS Services
Academic Institution	Arvind Gavali College of Engineering, Satara	Department of Computer Science & Engineering
Project Lead	Parth Shinde & Team	AI Architecture & Full-Stack Cloud Engineering

MISSION STATEMENT

Final Summary:
BIS Sahayak transforms dense government engineering standards into clear, spoken, bilingual guidance and instant license verification. By dismantling technical and language barriers, it empowers 1.4 billion citizens and millions of small businesses with product safety as an accessible national right.